Workflow labs use the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Compute a PCA, read a biplot, and explain loadings and scores without resorting to magical reasoning about "directions of variance".
- Distinguish PCA, factor analysis, canonical correlation analysis, and linear discriminant analysis by the question each answers.
- Fit an LDA classifier and interpret the discriminant coefficients.

Prerequisites

Matrix algebra (eigendecomposition); basic multivariate intuition.

Background

Dimension reduction is often described as "finding structure", but each of these methods has a concrete question behind it. PCA asks which directions in feature space have the largest variance; its axes are eigenvectors of the covariance matrix. Factor analysis asks which latent factors, plus unique variance per feature, reproduce the observed covariance; it is PCA's statistical cousin with a proper error model. Canonical correlation analysis asks which linear combinations of two feature sets are maximally correlated. Linear discriminant analysis asks which direction separates known classes while collapsing within-class variance.

They share machinery — all four involve eigenvalues of something — but they are not interchangeable. Using PCA where LDA was called for will waste the label information; using LDA where PCA was called for will produce a single, class-dependent axis and nothing else.

When you report a PCA, report the variance explained by each of the first few components and a biplot with named observations and loadings. When you report an LDA, report the classification accuracy on held-out data and the direction of each discriminant, not just the confusion matrix.

Setup

```
library(tidyverse)
library(MASS)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Use `iris` to illustrate PCA and LDA on the same data, compare their projections, and classify species with LDA.

2. Approach

Four morphometric variables, three species. PCA ignores the label; LDA uses it.

```
  pivot_longer(-Species) |>
  ggplot(aes(value, fill = Species)) +
  geom_density(alpha = 0.5) +
  facet_wrap(~ name, scales = "free") +
  labs(x = NULL, y = "density")
```

3. Execution

PCA on the four numeric features, then LDA.

```
summary(pca)$importance[, 1:3]

lda_fit <- lda(Species ~ ., data = iris)
lda_fit$scaling
```

```
scores_pca <- as_tibble(pca$x[, 1:2]) |> mutate(Species = iris$Species)
scores_lda <- as_tibble(predict(lda_fit)$x) |> mutate(Species = iris$Species)
p1 <- ggplot(scores_pca, aes(PC1, PC2, colour = Species)) +
  geom_point(alpha = 0.7) + labs(title = "PCA (unsupervised)")
p2 <- ggplot(scores_lda, aes(LD1, LD2, colour = Species)) +
  geom_point(alpha = 0.7) + labs(title = "LDA (uses labels)")
gridExtra::grid.arrange(p1, p2, ncol = 2)
```

4. Check

Hold-out accuracy for LDA (simple 70/30 split).

```
fit2 <- lda(Species ~ ., data = iris[idx, ])
pred <- predict(fit2, iris[-idx, ])$class
mean(pred == iris$Species[-idx])
```

5. Report

> On the *iris* data, the first two principal components explained `round(sum(summary(pca)$importance[2, 1:2]) * 100, 1)`% of the total variance. A linear discriminant classifier achieved hold-out accuracy of `round(mean(pred == iris$Species[-idx]) * 100, 1)`% on a 70/30 split.

In this low-dimensional problem, PCA already separates the species visually because the morphometric variables are correlated with species. In realistic biomedical data, that is rarely true: the supervised projection (LDA, or a classifier's decision function) is usually sharper.

Common pitfalls

- Reporting "PC1 is age" because age happens to correlate with it.
- Running LDA when classes are hugely imbalanced; use QDA or a proper classifier instead.
- Comparing variance-explained across datasets as if it were a quality score.

Further reading

- Jolliffe IT (2002), *Principal Component Analysis*, 2nd ed.
- Mardia, Kent, Bibby, *Multivariate Analysis*, chapters on LDA/CCA.

Session info

```
```

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)